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School of Computing

Informatics Institute of Technology (IIT Campus)

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Pseudocode for CRUD Operations

Program description:

The Task Management System is a simple Python application that allows users to manage their tasks effectively. The system supports basic CRUD (Create, Read, Update, and Delete) operations, enabling users to add, view, update, and remove tasks.

Steps of the Program (Pseudocode):

```
# PROGRAM Task Management System

START

    DECLARE tasks as an empty list

    FUNCTION add_task

        WHILE True DO

            PROMPT user to input task name

            IF task name is not empty OR task name is not just whitespace THEN

                BREAK

            ELSE

                PRINT "Task name cannot be empty. Please try again.

            END IF

        END WHILE


        PROMPT user to input task description


        WHILE True DO

            PROMPT user to input task priority

            priority = USER_INPUT("Enter task priority (High, Medium,
            Low)").capitalize()
```

```
        IF priority is in [“High”, “Medium”, “Low”] THEN
            BREAK
        ELSE
            PRINT "Invalid priority. Please choose between High, Medium,
            Low"
        END IF
    END WHILE


    WHILE True DO
        PROMPT user to input task due date (DD-MM-YYYY)
        due_date = USER INPUT(“Enter task Due Date (DD-MM-YYYY):  ”)
        TRY
            VALIDATE due_date using format DD-MM-YYYY
            BREAK
        CATCH ValueError
            PRINT "Invalid Input. Please enter the date in YYYY-MM-DD
            format."
    task = [ task_name, task_description, task_priority, task_due_date]
    ADD the task to the tasks
    PRINT "Task added successfully!"
END FUNCTION


FUNCTION view_tasks
    IF tasks is empty THEN
        PRINT "No tasks available."
```

```
        ELSE
            FOR each task in tasks DO
                PRINT task details (task name, priority, due date, description)
            END FOR
        END IF
END FUNCTION


FUNCTION update_task
        IF tasks is empty THEN
            PRINT "No tasks to update."
        RETURN
        END IF
        PROMPT user to input task number to update
        TRY
            task_id = USER_INPUT() - 1
            IF 0 <= task_id < length of tasks
            PROMPT user to input new task name (or keep the old one if empty)
            PROMPT user to input new task description (or keep the old one if
            empty)
            WHILE True DO
                PROMPT user to input new task priority (or keep the old one if
                empty)
                task_priority = USER_INPUT().capitalize()
                IF task_priority is empty
                    BREAK
```

```
        ELIF task_priority IS in ["High", "Medium", "Low"]
            task[2] = task_priority
        ELSE
            PRINT"Invalid priority. Please choose from High,
            Medium, or Low."
        END IF
    END WHILE
    WHILE True DO
        PROMPT user to input new task due date (or keep the old one
        if empty)
        due_date = USE_INPUT()
        IF due_date is empty
            BREAK
        TRY
            VALIDATE due_date  using format DD-MM-YYYY
            task[3] =due_date
            BREAK
        CATCH ValueError
            PRINT "Invalid date format, Please enter the date in
            DD-MM-YYYY"
    END WHILE
    PRINT "Task updated!"
    ELSE
        PRINT "Invalid task number."
    END IF
```

```
        CATCH ValueError
            PRINT "Invalid Input!"
END FUNCTION


FUNCTION delete_task
        IF tasks is empty THEN
            PRINT "No tasks to delete."
        RETURN
        END IF
        PROMPT user to input task number to delete
        TRY
            task_id = USER_INPUT() - 1
            IF 0 <= task_id < length of tasks
                REMOVE the task from the tasks list
                PRINT "Task deleted!"
            ELSE
                PRINT "Invalid task number."
            END IF
        CATCH ValueError
            PRINT “Invalid Input!”
END FUNCTION


FUNCTION main
        WHILE True DO
```

```
                PRINT task manager menu options
                PROMPT user to choose an option
                IF the choice is '1' THEN
                        CALL add_task
                ELSE IF the choice is '2' THEN
                        CALL view_tasks
                ELSE IF the choice is '3' THEN
                        CALL update_task
                ELSE IF the choice is '4' THEN
                        CALL delete_task
                ELSE IF the choice is '5' THEN
                        PRINT "Goodbye!" and BREAK the loop
                ELSE
                PRINT "Invalid option! Please select a valid option."
                END IF
            END WHILE
        END FUNCTION

        CALL main to start the program
END
```

Test Cases

Add a Task

Test Case: Valid Inputs

Test Case ID	Input	Expected Result	Actual Result	Pass / Fail
TC_01	"Study Mathematics", "For upcoming exams.", "High", 28-03-2025	Task 'Study Mathematics' added successfully! Task saved to file.	Task 'Study Mathematics' added successfully! Task saved to file.	Pass

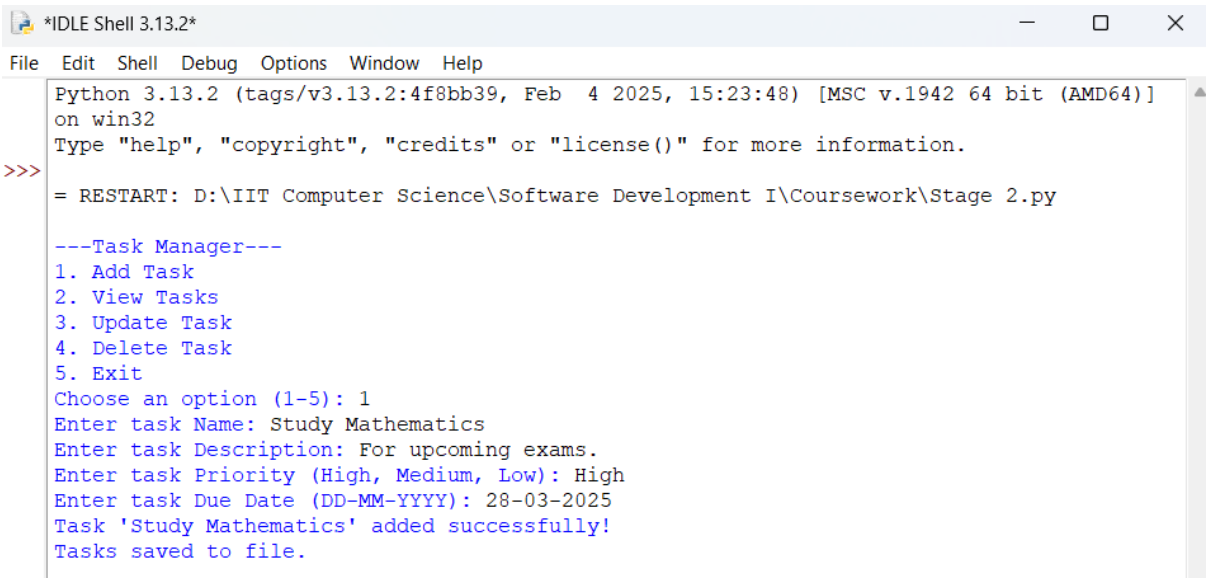


Figure 1: Test Case 1

Test Case: Empty task name

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_02	-	Task name cannot be empty. Please try again.	Ensures that the application does not allow a task with an empty list.	Task name cannot be empty. Please try again.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======
No previous tasks found. Starting with an empty task list.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name:
Task name cannot be empty. Please try again.
Enter task Name:
```

Figure 2: Test Case 2

Test Case: Invalid Priority

Test Case ID	Inputs	Expected Result	Description	Actual Result	Pass / Fail
TC_03	Exam	Invalid priority. Please choose between High, Medium, or Low.	Verifies that the system rejects an invalid priority value that is not "High", "Medium", or "Low"	Invalid priority. Please choose between High, Medium, or Low.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Study Mathematics
Enter task Description: For upcoming exams
Enter task Priority (High, Medium, Low): Exam
Invalid priority. Please choose from High, Medium, or Low.
```

Figure 3: Test Case 3

Test Case: Invalid Date

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_04	2025-03-28	Invalid date format. Please enter the date in DD-MM-YYYY format.	Check whether the system does not accept dates in an incorrect format.	Invalid date format. Please enter the date in DD-MM-YYYY format.	Pass
TC_05	Five				Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 1 (w2151928).
py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Enter task Name: Study Mathematics
Enter task Description: For upcoming exams.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 2025-03-28
Invalid Input. Please enter the date in DD-MM-YYYY format.
```

Figure 4: Test Case 4

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 1 (w2151928).
py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Enter task Name: Study Mathematics
Enter task Description: For upcoming exams.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): Five
Invalid Input. Please enter the date in DD-MM-YYYY format.
```

Figure 5: Test Case 5

View Tasks

Test Case: No tasks in the list.

Test Case ID	Pre-Condition	Expected Result	Actual Result	Pass / Fail
TC_06	No tasks added	No tasks available.	No tasks available.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======
No previous tasks found. Starting with an empty task list.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 2
No tasks available.
```

Figure 6: Test Case 6

Test Case: Existing Tasks

Test Case ID	Pre-Condition	Expected Result	Actual Result	Pass / Fail
TC_07	Tasks exist	Task list displayed.	Task list displayed.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 2.py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 2
Task 1
  Task Name: Study Mathematics
  Task Priority: High
  Due Date: 28-03-2025
  Task Description: For upcoming exams.
Task 2
  Task Name: Study CSF
  Task Priority: High
  Due Date: 10-03-2025
  Task Description: For upcoming exams.
```

Figure 7: Test Case 7

Update Tasks

Test Case: Updating an existing task

Test Case ID	Inputs	Expected Result	Actual Result	Pass / Fail
TC_08	Enter task number to update: "2" Name (Study CSF): “Practice Chess” Description (For upcoming exams.): “For an upcoming tournament.” Priority (High): "Medium" Due date (10-03-2025): " 26-05-2025 "	Task updated! Tasks saved to file.	Task updated! Tasks saved to file.	Pass

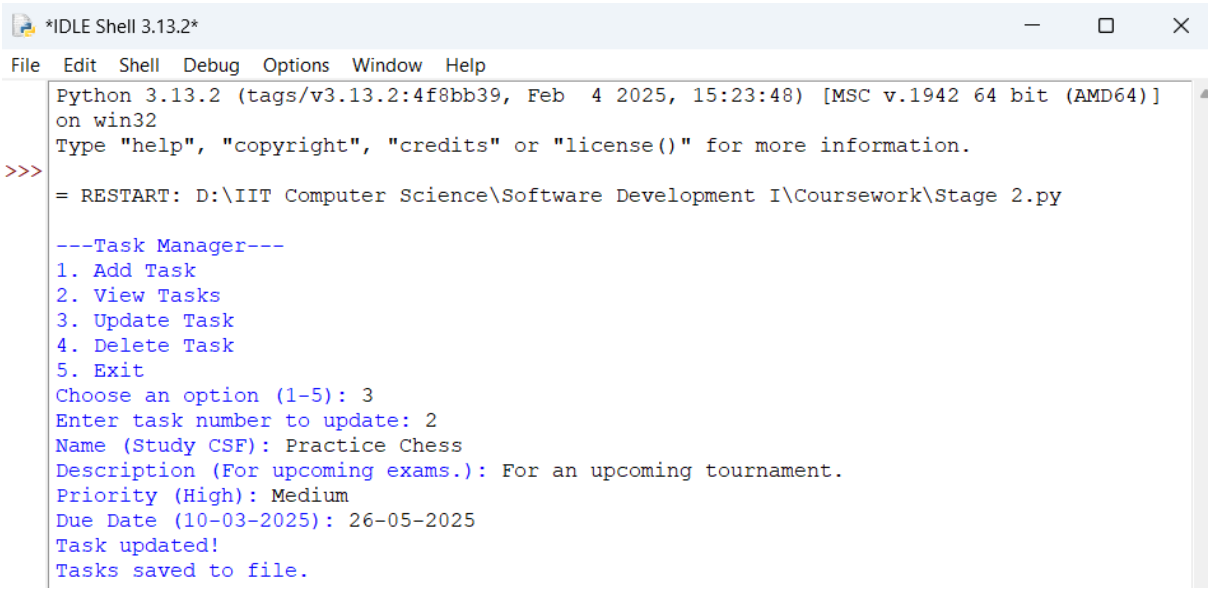


Figure 8: Test Case 8

Test Case: Invalid task number

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_09	4	Task not found.	Tests non-existing task name	Task not found.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 4
Invalid task number!
```

Figure 9: Test Case 9

Test Case: Invalid Priority

Test Case ID	Inputs	Expected Result	Description	Actual Result	Pass / Fail
TC_10	L	Invalid priority. Please choose between High, Medium, or Low.	Verifies that the system rejects an invalid priority value that is not "High", "Medium", or "Low"	Invalid priority. Please choose between High, Medium, or Low.	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 1
Name (Study Mathematic): Practice Chess
Description (For upcoming exams.): For an upcoming tournament.
Priority (High): L
Invalid priority. Please choose between High, Medium, or Low.
```

Figure 10: Test Case 10

Test Case: Invalid Date

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_11	2025-03-28	Invalid date format. Please enter the date in DD-MM-YYYY format.	Check whether the system does not accept dates in an incorrect format.	Invalid date format. Please enter the date in DD-MM-YYYY format.	Pass
TC_12	Five				Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 1
Name (Study Mathematic): Practice Chess
Description (For upcoming exams.): For an upcoming tournament.
Priority (High): Medium
Due Date (28-03-2025): 2025-03-28
Invalid date format, Please enter the date in DD-MM-YYYY
```

Figure 11: Test Case 11

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 1
Name (Study Mathematic): Practice Chess
Description (For upcoming exams.): For an upcoming tournament.
Priority (High): Medium
Due Date (28-03-2025): Five
```

Figure 12: Test Case 12

Delete Tasks

Test Case: No existing tasks in the list.

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_13	4	No tasks to delete.	Attempt to delete when no tasks are available	No tasks to delete.	Pass

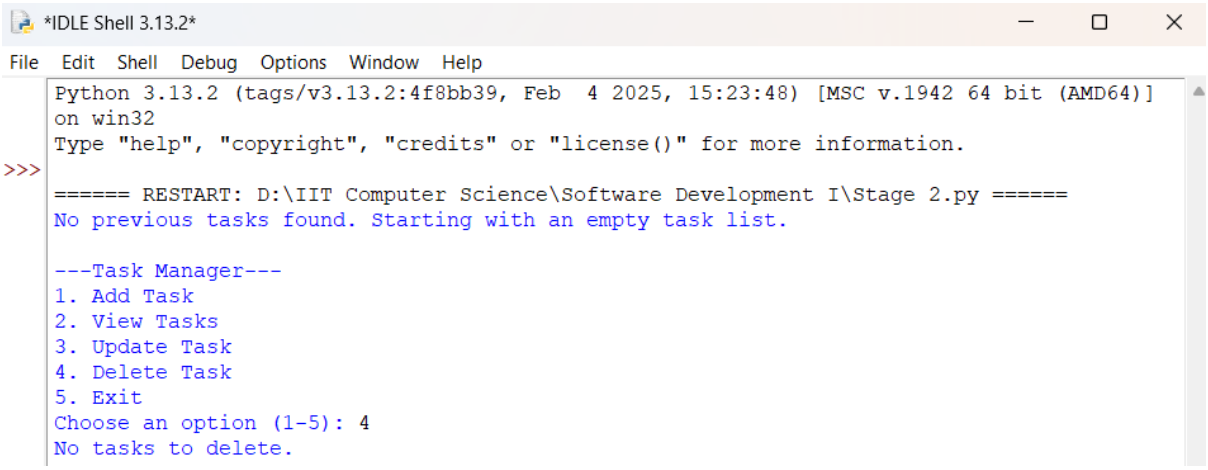


Figure 13: Test Case 13

Test Case: Valid deletion of task

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_14	Enter task number to delete: “1”	Task 'Study Mathematics' deleted!	Tests valid deletion of an existing task.	Task 'Study Mathematics' deleted!	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 2.py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 4
Enter task number to delete: 1
Task 'Study Mathematics' deleted!
Tasks saved to file.
```

Figure 14: Test Case 14

Test Case: Invalid Task Number

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC_15	3	Invalid task number!	Attempt to delete when no tasks are available	Invalid task number!	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: D:\IIT Computer Science\Software Development I\Stage 2.py ======

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 4
Enter task number to delete: 3
Invalid task number!
```

Figure 15: Test Case 15

Test Cases for Text File Handling

Test Case: Loading Tasks from Text File

Test Case ID	Steps	Expected Result	Pass / Fail
TC_16	Create a valid Tasks.txt file with tasks saved in the correct format.	The tasks in the file should be correctly loaded into the tasks list, and each task's details should be correctly displayed	Pass

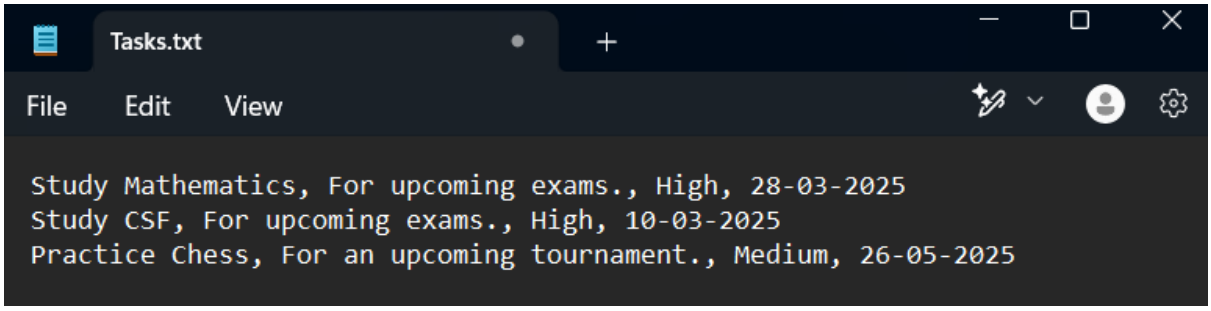


Figure 16: Test Case 16.0

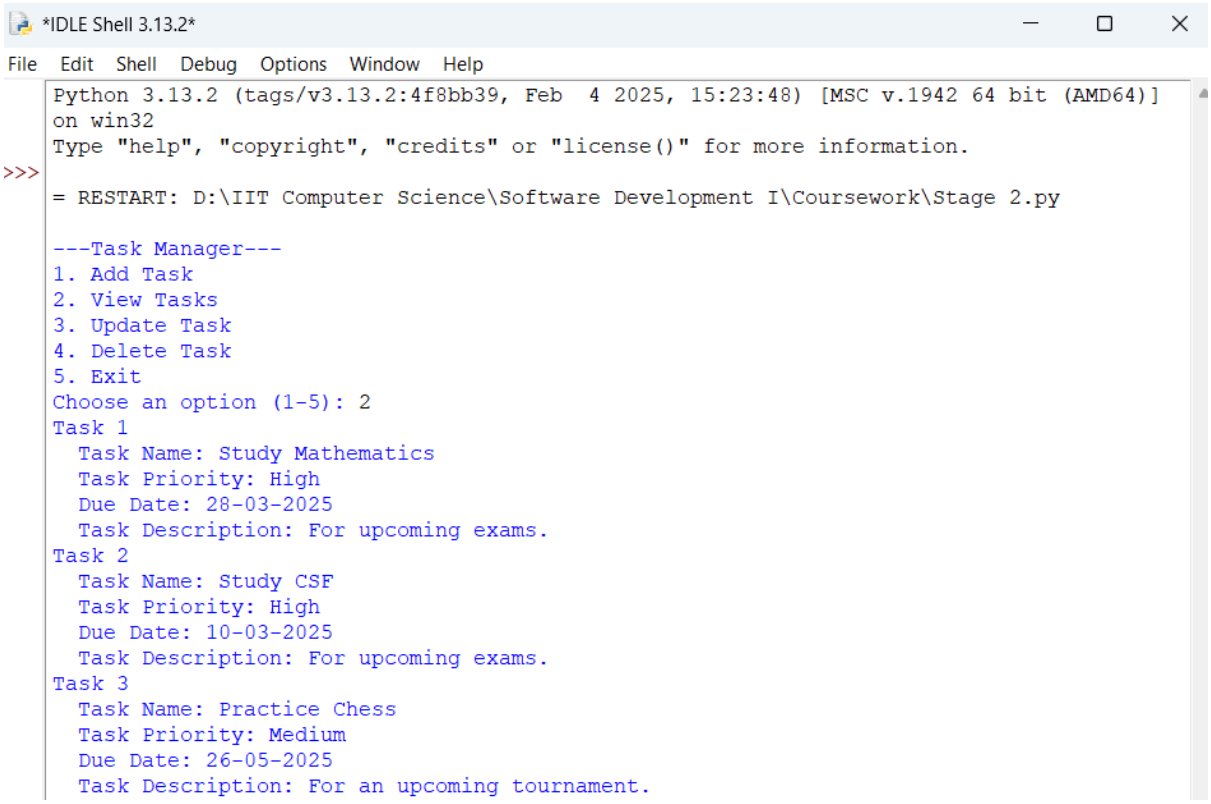


Figure 17: Test Case 16.1

Test Case ID	Steps	Expected Result	Pass / Fail
TC_17	Create a valid Tasks.txt file with tasks saved in the correct format.	The tasks in the file should be correctly loaded into the tasks list, and each task's details should be correctly displayed	Pass

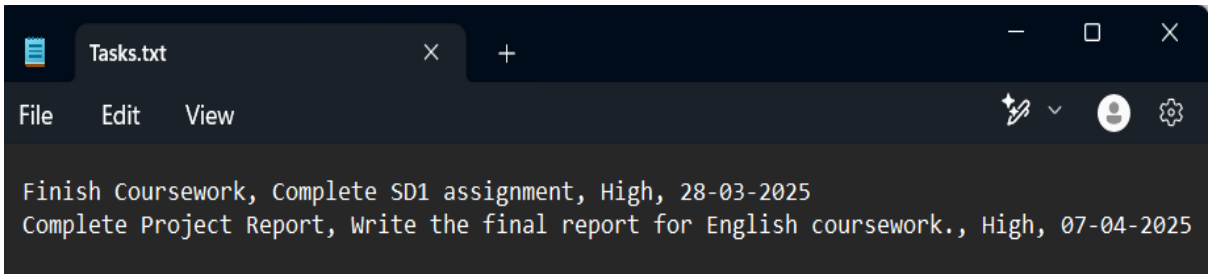


Figure 18: Test Case 17.0

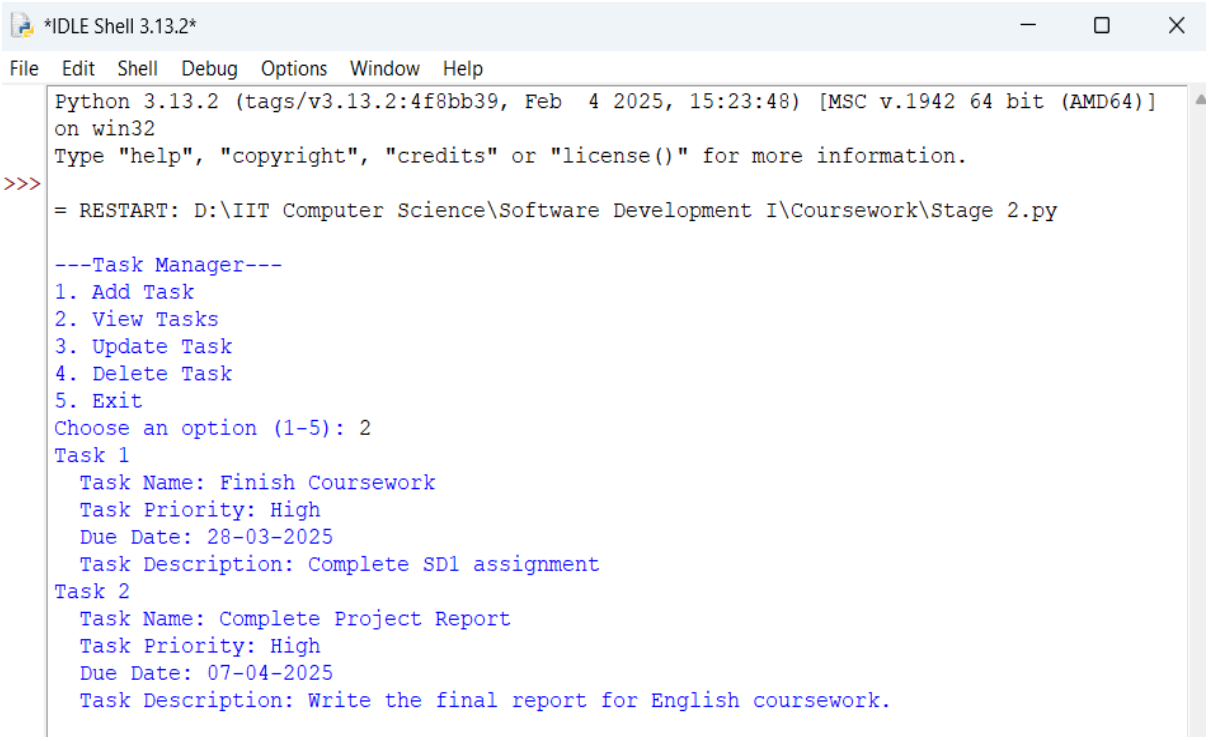


Figure 19: Test Case 17.1

Test Case: Saving Tasks to Text File

Test Case ID	Steps	Expected Result	Pass / Fail
TC_18	Add multiple tasks e.g.01, 3 tasks with different names, descriptions, priorities, and due dates.	All tasks should be saved in the file.	Pass
TC_19	e.g.02, 2 tasks with different names, descriptions, priorities, and due dates.		

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 2.py
No previous tasks found. Starting with an empty task list.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Study Mathematics
Enter task Description: For upcoming exams.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 28-03-2025
Task 'Study Mathematics' added successfully!
Tasks saved to file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Study CSF
Enter task Description: For upcoming exams.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 10-03-2025
Task 'Study CSF' added successfully!
Tasks saved to file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Practice Chess
Enter task Description: For an upcoming tournament.
Enter task Priority (High, Medium, Low): Medium
Enter task Due Date (DD-MM-YYYY): 26-05-2025
Task 'Practice Chess' added successfully!
Tasks saved to file.
```

Figure 20: Test Case 18.0

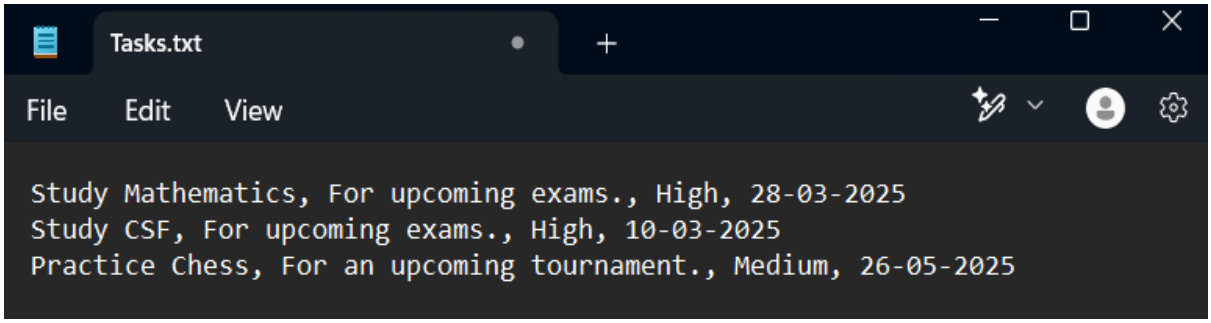


Figure 21: Test Case 18.1

Figure 22: Test Case 19.0

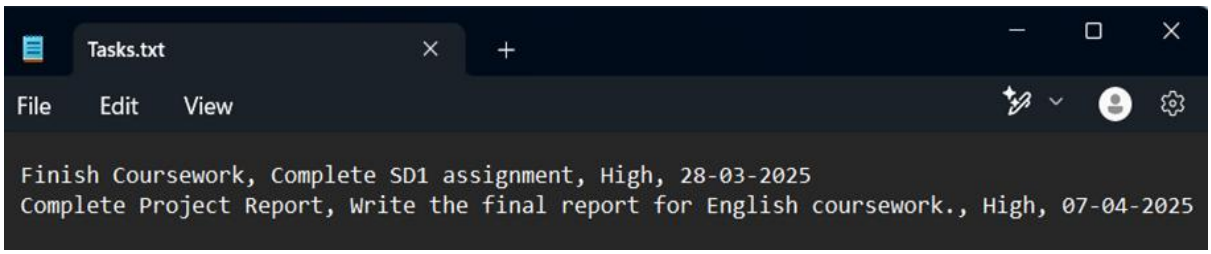


Figure 23: Test Case 19.1

Test Case: Handling Missing Text File

Test Case ID	Steps	Expected Result	Actual Result	Pass / Fail
TC_20	Run the program without "Task.txt" file.	No previous tasks found. Starting with an empty task list.	No previous tasks found. Starting with an empty task list.	Pass

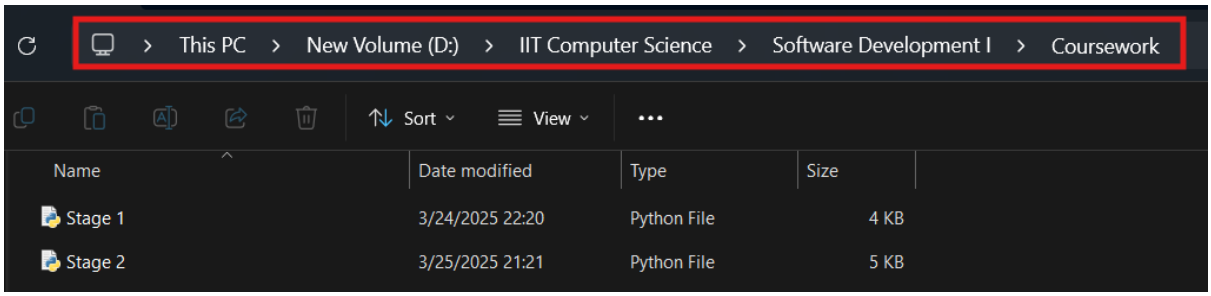


Figure 24: Test Case 20.0

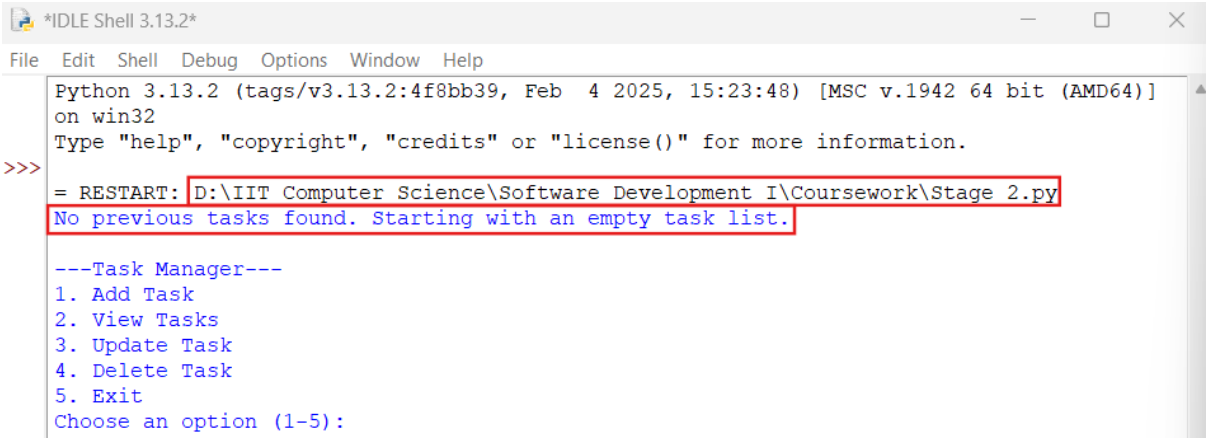


Figure 25: Test Case 20.1

Test Case: Text File Update

Test Case ID	Input	Expected Result	Actual Result	Pass / Fail
TC_21	Enter task number to update: "1"	Task updated!	Task updated!	Pass

```
*IDLE Shell 3.13.3*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Codes\Stage 2 (w2151928).py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 1
Name (Study Mathematics): Finish Coursework
Description (For upcoming exams.): Complete SD1 assignment.
Priority (High): High
Due Date (28-03-2025): 31-03-2025
Task updated!
Tasks saved to file.
```

Figure 26: Test Case 21.0

Before:

```
Tasks.txt
File  Edit  View

Study Mathematics, For upcoming exams., High, 28-03-2025
Study CSF, For upcoming exams., High, 10-03-2025
Practice Chess, For an upcoming tournament., Medium, 26-05-2025
```

Figure 27: Test Case 21.1

After:

```
Tasks.txt
File  Edit  View

Finish Coursework, Complete SD1 assignment., High, 31-03-2025
Study CSF, For upcoming exams., High, 10-03-2025
Practice Chess, For an upcoming tournament., Medium, 26-05-2025
```

Figure 28: Test Case 21.2

Test Case: Task Deletion (Text File)

Test Case ID	Input	Expected Result	Actual Result	Pass / Fail
TC_22	Enter task number to delete: “2”	Task 'Study CSF' deleted!	Task 'Study CSF' deleted!	Pass

```
*IDLE Shell 3.13.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Stage 2.py

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 4
Enter task number to delete: 2
Task 'Study CSF' deleted!
Tasks saved to file.
```

Figure 29: Test Case 22.0

Before:

```
Tasks.txt
File  Edit  View

Study Mathematics, For upcoming exams., High, 28-03-2025
Study CSF, For upcoming exams., High, 10-03-2025
Practice Chess, For an upcoming tournament., Medium, 26-05-2025
```

Figure 30: Test Case 22.1

After:

```
Tasks.txt
File  Edit  View

Study Mathematics, For upcoming exams., High, 28-03-2025
Practice Chess, For an upcoming tournament, Medium, 26-05-2025
```

Figure 31: Test Case 22.2

Test Cases for JSON File Handling

Test Case: Load Task from JSON

Test Case ID	Steps	Expected Result	Pass/Fail
TC_23	Create a valid Tasks.json file with tasks saved in the correct format.	The tasks in the file should be correctly loaded into the tasks list, and each task's details should be correctly displayed.	Pass

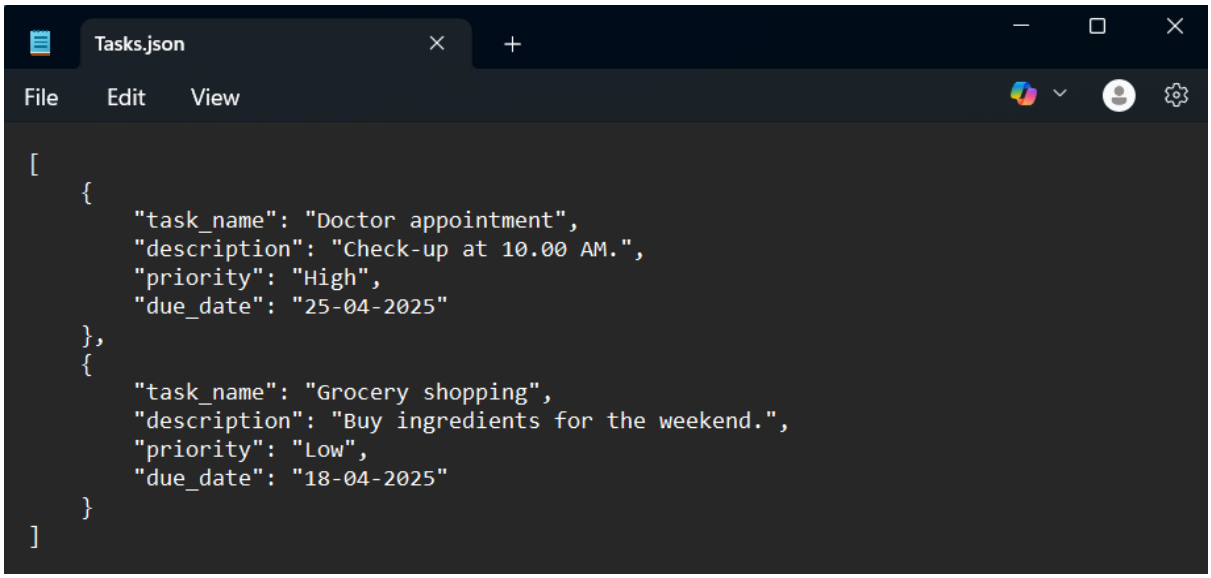


Figure 32: Test Case 23.0

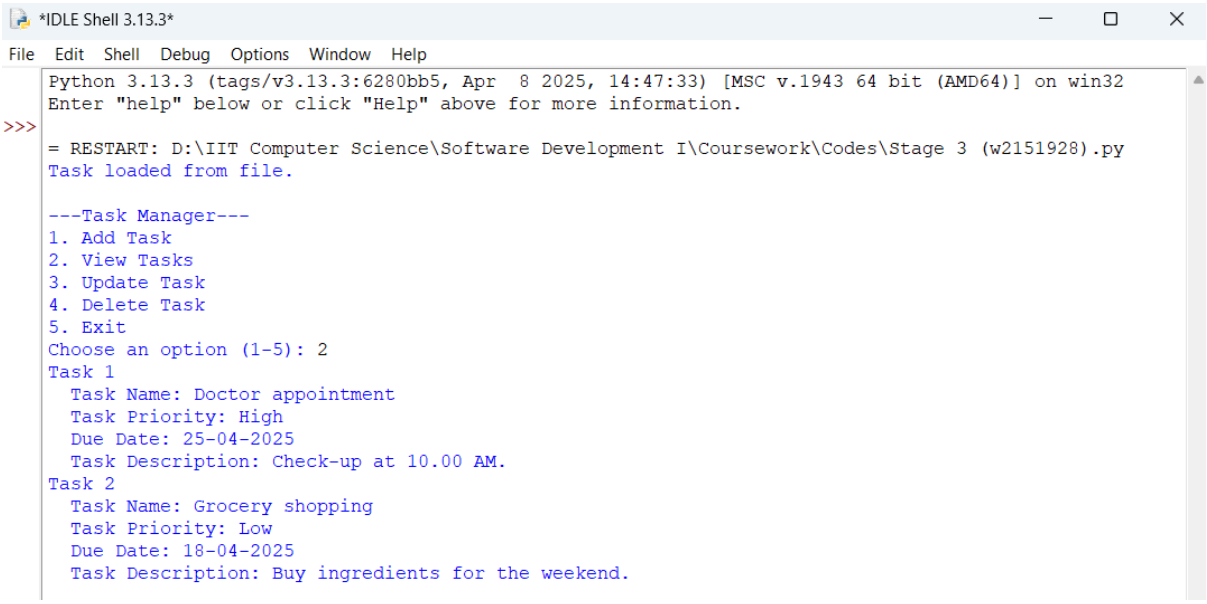


Figure 33: Test Case 23.1

Test Case ID	Steps	Expected Result	Pass/Fail
TC_24	Create a valid Tasks.json file with tasks saved in the correct format.	The tasks in the file should be correctly loaded into the tasks list, and each task's details should be correctly displayed.	Pass

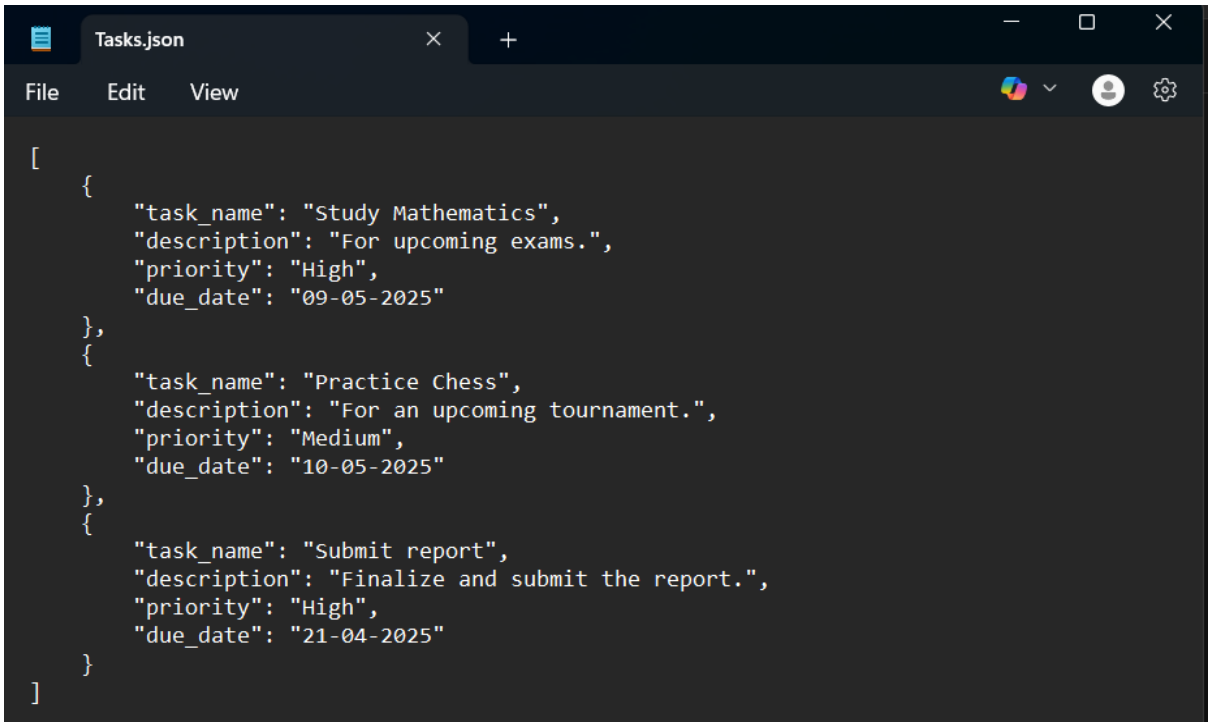


Figure 34: Test Case 24.0

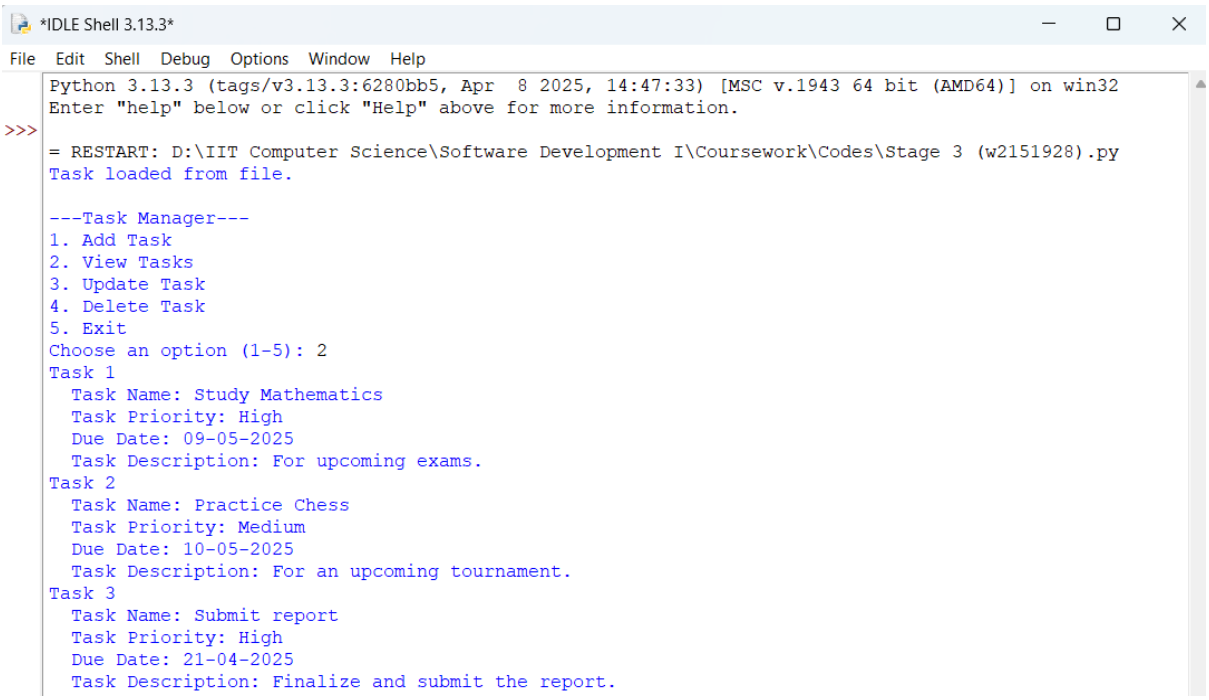


Figure 35: Test Case 24.1

Test Case: Save Tasks to JSON

Test Case ID	Steps	Expected Result	Pass / Fail
TC_25	Add multiple tasks e.g.01, 3 tasks with different names, descriptions, priorities, and due dates.	All tasks should be saved in the file.	Pass
TC_26	e.g.02, 2 tasks with different names, descriptions, priorities, and due dates.		

```
*IDLE Shell 3.13.3*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Codes\Stage 3 (w2151928).py
No previous tasks found. Starting with an empty task list.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Study Mathematics
Enter task Description: For upcoming exams.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 09-05-2025
Task 'Study Mathematics' added successfully!
Tasks saved to file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Practice Chess
Enter task Description: For an upcoming tournament.
Enter task Priority (High, Medium, Low): Medium
Enter task Due Date (DD-MM-YYYY): 10-05-2025
Task 'Practice Chess' added successfully!
Tasks saved to file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Submit report
Enter task Description: Finalize and submit the report.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 21-04-2025
Task 'Submit report' added successfully!
Tasks saved to file.
```

Figure 36: Test Case 25.0

Tasks.json

File Edit View


```
[
    {
        "task_name": "Study Mathematics",
        "description": "For upcoming exams.",
        "priority": "High",
        "due_date": "09-05-2025"
    },
    {
        "task_name": "Practice Chess",
        "description": "For an upcoming tournament.",
        "priority": "Medium",
        "due_date": "10-05-2025"
    },
    {
        "task_name": "Submit report",
        "description": "Finalize and submit the report.",
        "priority": "High",
        "due_date": "21-04-2025"
    }
]
```


Figure 37: Test Case 25.1

IDLE Shell 3.13.3

File Edit Shell Debug Options Window Help

```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Codes\Stage 3 (w2151928).py
No previous tasks found. Starting with an empty task list.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Doctor appointment
Enter task Description: Check-up at 10.00 AM.
Enter task Priority (High, Medium, Low): High
Enter task Due Date (DD-MM-YYYY): 25-04-2025
Task 'Doctor appointment' added successfully!
Tasks saved to file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 1
Enter task Name: Grocery shopping
Enter task Description: Buy ingredients for the weekend.
Enter task Priority (High, Medium, Low): Low
Enter task Due Date (DD-MM-YYYY): 18-04-2025
Task 'Grocery shopping' added successfully!
Tasks saved to file.
```

Figure 38: Test Case 26.0

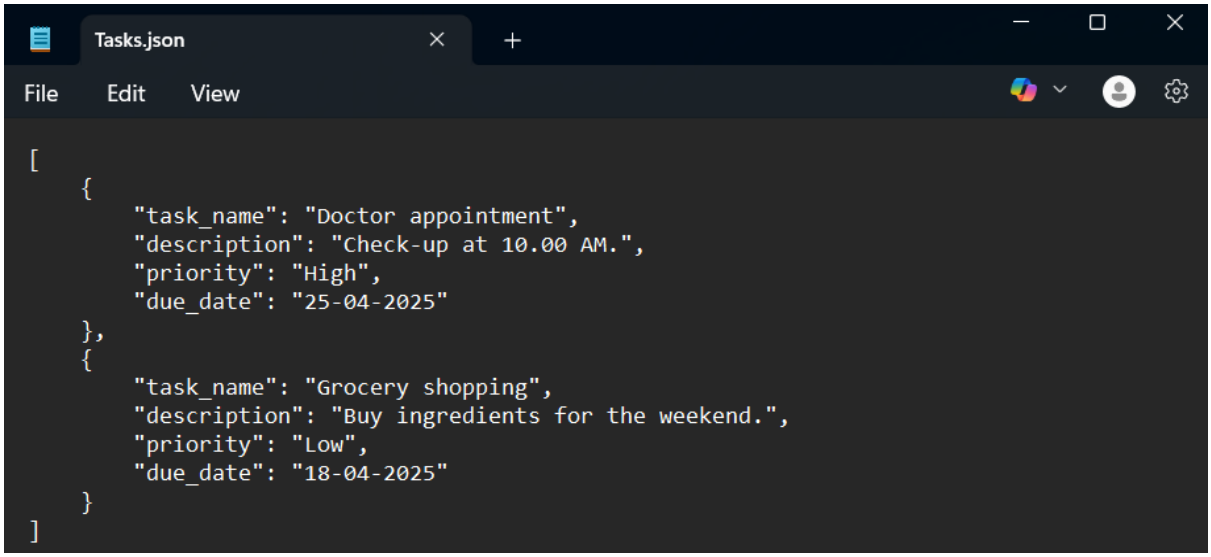


Figure 39: Test Case 26.1

Test Case: Handling Missing JSON File

Test Case ID	Steps	Expected Result	Actual Result	Pass / Fail
TC_27	Run the program without "Task.json" file.	No previous tasks found. Starting with an empty task list.	No previous tasks found. Starting with an empty task list.	Pass

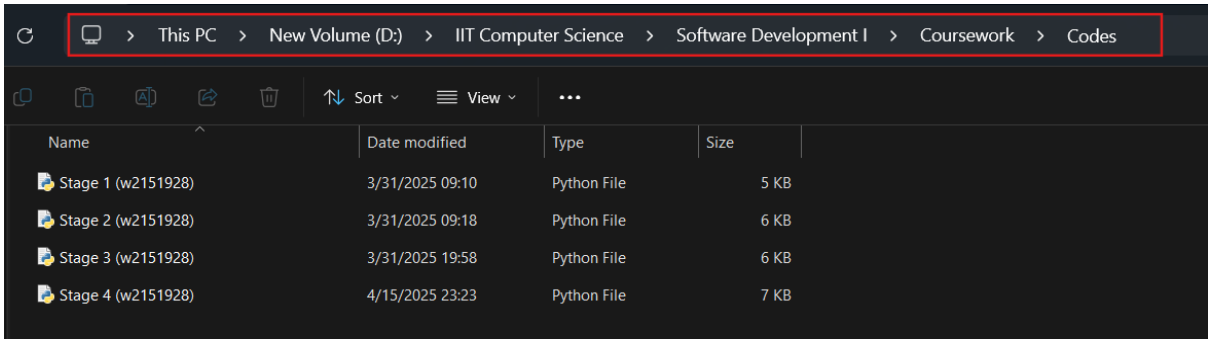


Figure 40: Test Case 27.0

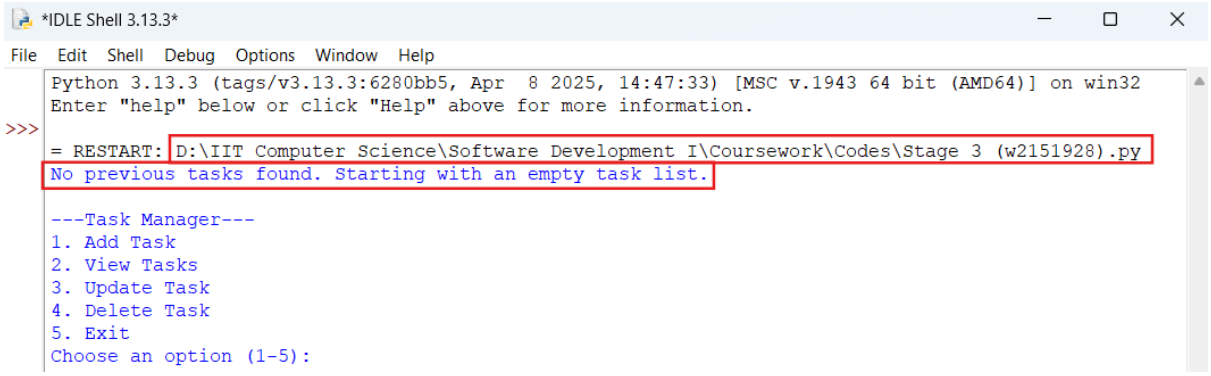


Figure 41: Test Case 27.1

Test Case: JSON File Update

Test Case ID	Input	Expected Result	Actual Result	Pass / Fail
TC_28	Enter task number to update: “2”	Task updated!	Task updated!	Pass

```
*IDLE Shell 3.13.3*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: D:\IIT Computer Science\Software Development I\Coursework\Codes\Stage 3 (w2151928).py
Task loaded from file.

---Task Manager---
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option (1-5): 3
Enter task number to update: 2
Name (Practice Chess): Doctor appointment
Description (For an upcoming tournament.): Check-up at 10.00 AM.
Priority (Medium): High
Due Date (10-05-2025): 25-04-2025
Task updated!
Tasks saved to file.
```

Figure 42: Test Case 28.0

Before:

```
Tasks.json
File  Edit  View
[
    {
        "task_name": "Study Mathematics",
        "description": "For upcoming exams.",
        "priority": "High",
        "due_date": "09-05-2025"
    },
    {
        "task_name": "Practice Chess",
        "description": "For an upcoming tournament.",
        "priority": "Medium",
        "due_date": "10-05-2025"
    },
    {
        "task_name": "Submit report",
        "description": "Finalize and submit the report.",
        "priority": "High",
        "due_date": "21-04-2025"
    }
]
```

Figure 43: Test Case 28.1

After:

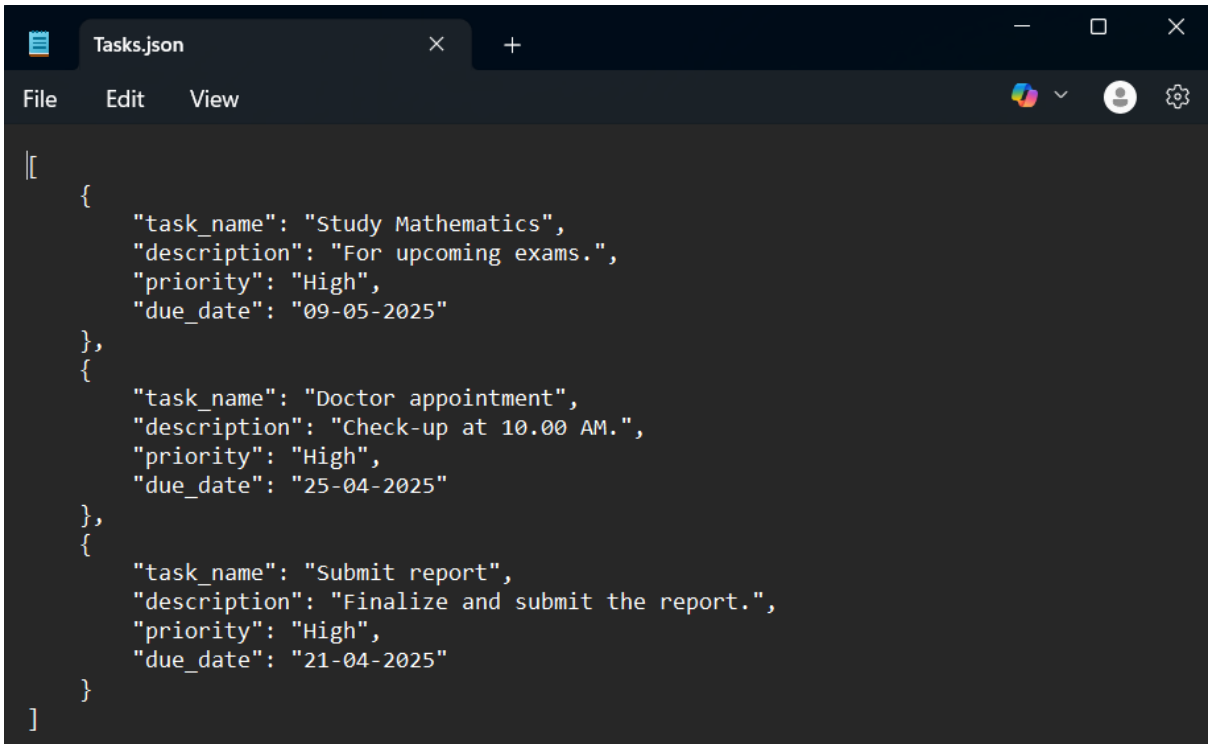


Figure 44: Test Case 28.2

Test Case: Task Deletion (JSON)

Test Case ID	Input	Expected Result	Actual Result	Pass / Fail
TC_29	Enter task number to delete: “3”	Task 'Submit report' deleted!	Task 'Submit report' deleted!	Pass

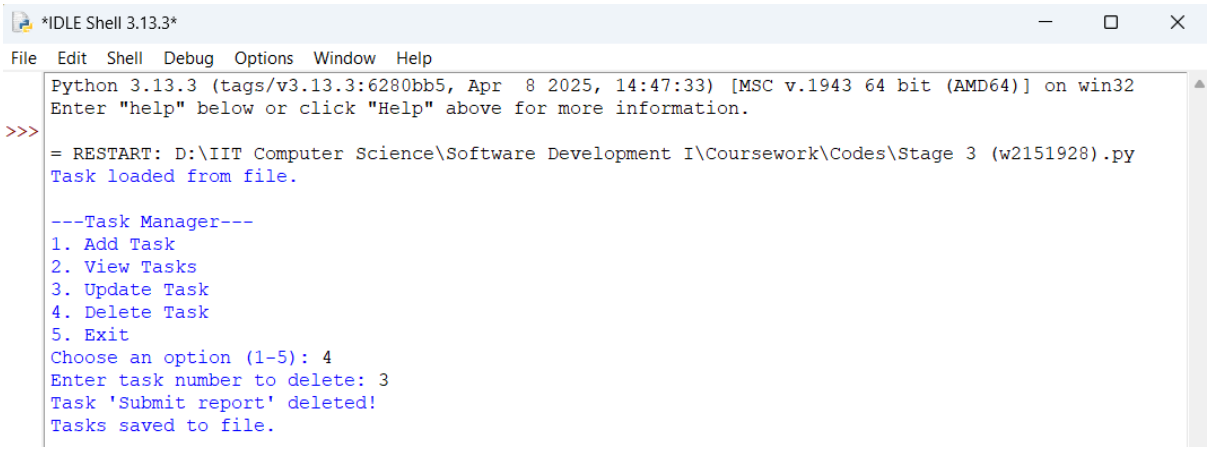
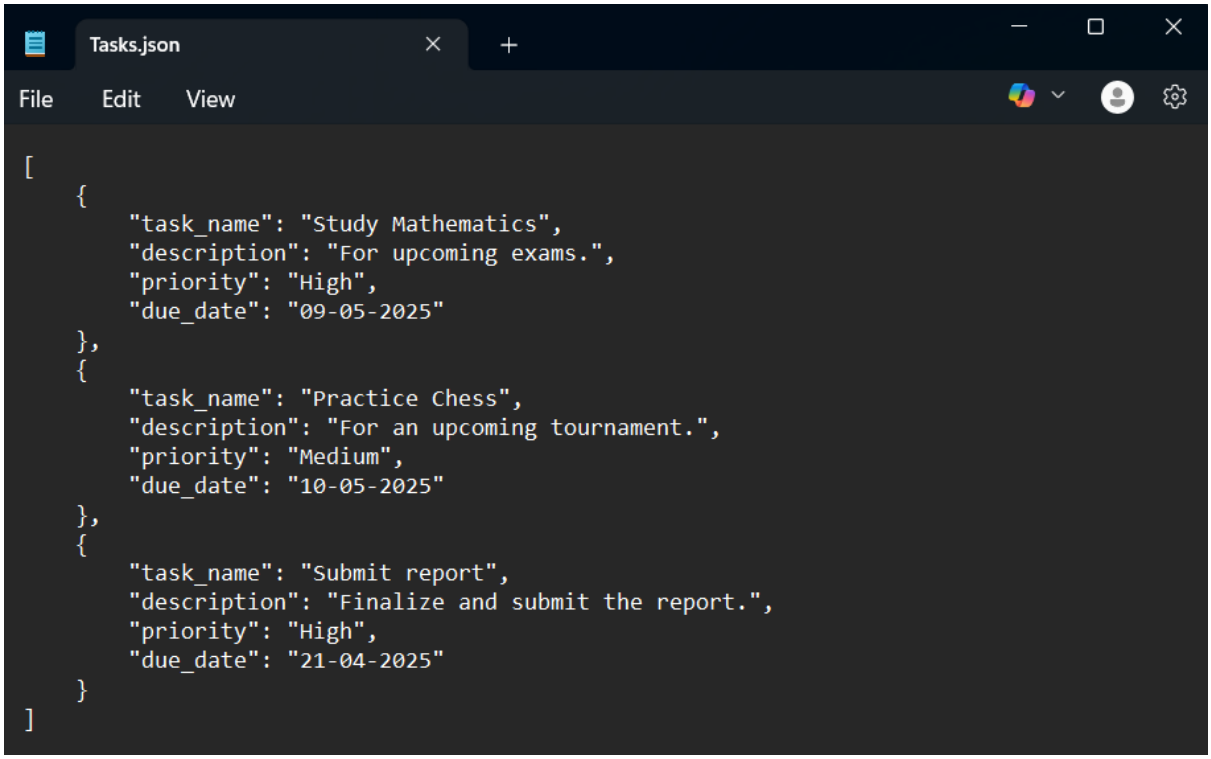


Figure 45: Test Case 29.0

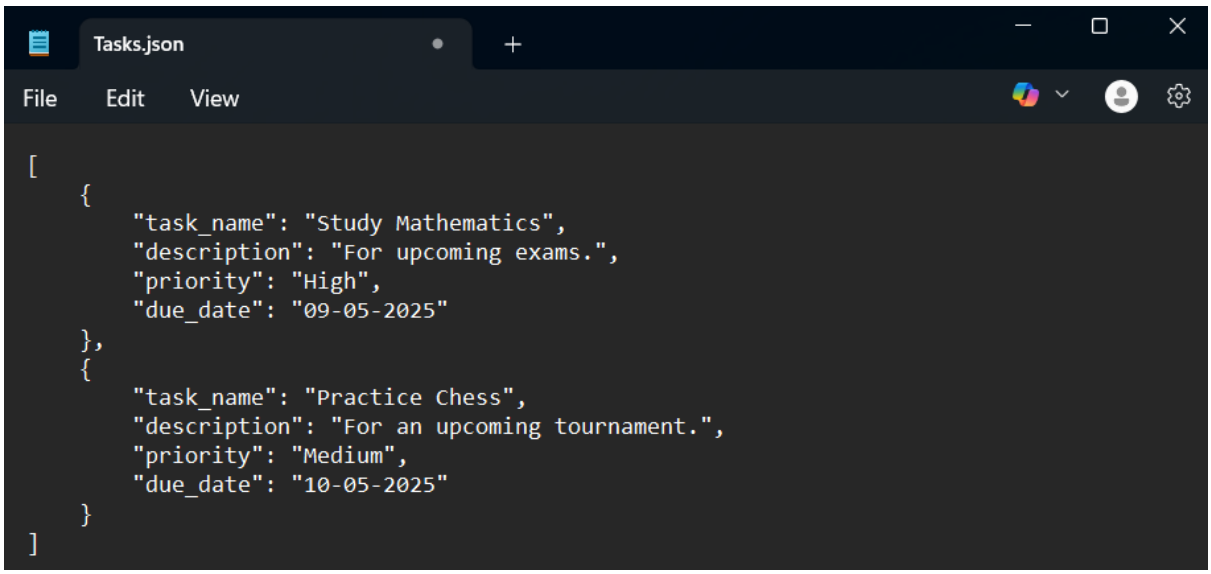
Before:




```
[
    {
        "task_name": "Study Mathematics",
        "description": "For upcoming exams.",
        "priority": "High",
        "due_date": "09-05-2025"
    },
    {
        "task_name": "Practice Chess",
        "description": "For an upcoming tournament.",
        "priority": "Medium",
        "due_date": "10-05-2025"
    },
    {
        "task_name": "Submit report",
        "description": "Finalize and submit the report.",
        "priority": "High",
        "due_date": "21-04-2025"
    }
]
```


Figure 46: Test Case 29.1

After:




```
[
    {
        "task_name": "Study Mathematics",
        "description": "For upcoming exams.",
        "priority": "High",
        "due_date": "09-05-2025"
    },
    {
        "task_name": "Practice Chess",
        "description": "For an upcoming tournament.",
        "priority": "Medium",
        "due_date": "10-05-2025"
    }
]
```


Figure 47: Test Case 29.2

Graphical User Interface Documentation and User Instructions

GUI Explanation

The **Personal Task Manager** application is a user-friendly desktop application built with Python and Tkinter. It helps users to view, filter, and sort tasks easily using an intuitive tabular interface.

GUI Components

Components Description

1. **Title Bar**
    - **Label:** Displays the name of the application – "**Personal Task Manager**"
2. **Search and Filter Section**
    - **Task Name Entry Field:** Enter a keyword to search tasks by name.
    - **Priority Dropdown menu:** Select from "All", "High", "Medium", or "Low" to filter tasks based on priority.
    - **Due Date Entry Field:** enter a specific due date in "DD-MM-YYYY" format to filter task due on that date.
    - **Filter Button:** Apply the filter criteria based on the inputs above.
3. **Task Display Table**
    - Implemented using a **Treeview** widget.
    - Display all tasks in a tabular format with the following columns:
        - **Name**
        - **Description**
        - **Priority**
        - **Due Date**
4. **Column Sorting**
    - Each column header is clickable.
    - Clicking a header sorts tasks based on that column in ascending or descending order.
5. **Responsive Design**
    - The Treeview table is scrollable and resizes with the window, making the interface flexible and user-friendly.

Graphical User Interface (Screenshot)

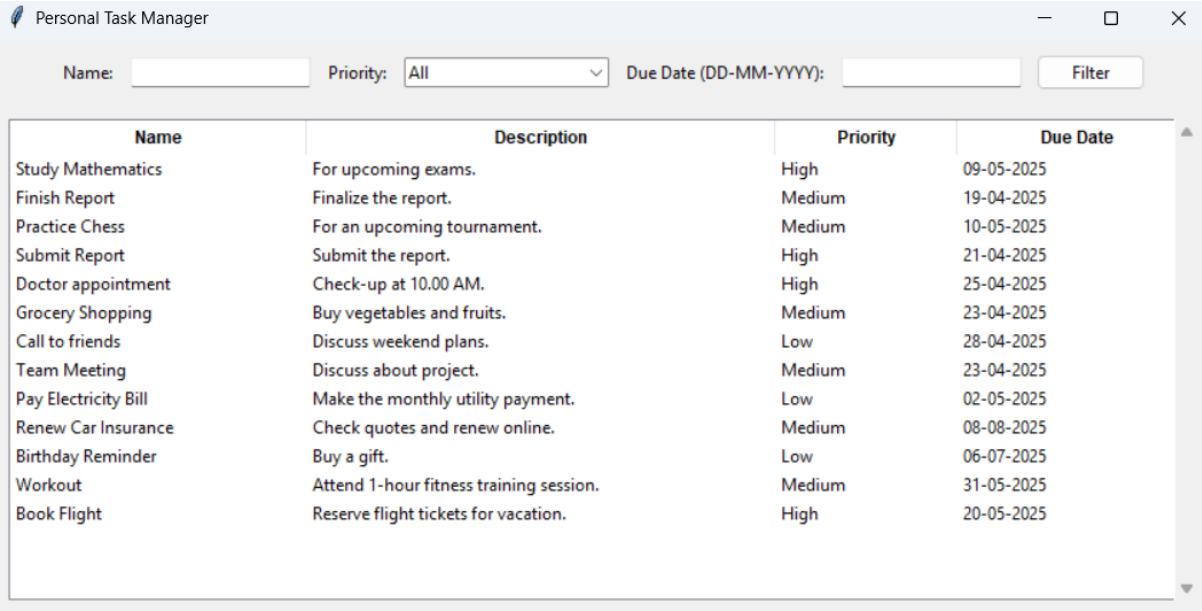


Figure 48: GUI

Instructions for Search and Sort Functionalities

Step-by-Step Usage Guide

1. **Filter tasks (Search)**

   Steps:

   1. **Start the application:**
      - Run the Python file (e.g., Stage 4(w2151928))
      - The GUI window will appear.
   2. **The top section, use any of the following filters:**
      - **Task name:** enter any part of the task name (e.g., “Submit report”)
      - **Priority:** choose from “All”, “High”, “Medium”, or “Low”.
      - **Due Date (DD-MM-YYYY):** Enter a full due date in DD-MM-YYYY format (e.g., 21-04-2025)
   3. **Click the “filter" button.**
   4. **This table updates to show only matching tasks.**

2. **Sorting Tasks**

   Can sort tasks by clicking any column header:

   - Click Name, Description, Priority, Due Date (DD-MM-YYYY) headers.
   - The first click sorts ascending, and the second click sorts descending.

3. **Clear filters**
    - To clear all filters, remove all text from entry fields and set priority to "All", then click **Filter** again.

Tips for Best User Experience

- Use partial task names to find multiple relevant tasks (e.g., typing "report" finds "Finish Report" and "Submit Report")
- When filtering by **date**, double-check the format (DD-MM-YYYY) to avoid mismatches.
- Use the column sort feature to quickly prioritize what matters most (e.g., sort by due date to find the nearest deadline).
- Use filters first, then click a column header to sort the filtered results.
- Combine filters (e.g., name + priority) for more targeted searches.

Test Cases for GUI

Test Case: Filter Tasks

Task ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_30	Enter Name: "Report" and click the Filter button.	Only relevant data is visible.	Only relevant data is visible.	Pass
TC_31	Enter Priority: "High" and click the Filter button.	Only relevant data is visible.	Only relevant data is visible.	Pass
TC_32	Enter Due Date: "23-04-2025" and click the Filter button.	Only relevant data is visible.	Only relevant data is visible.	Pass
TC_33	Enter Name: "Submit Report", Priority: "High", Due Date (DD-MM-YYYY): "21-04-2025" and click the Filter button.	Only relevant data is visible.	Only relevant data is visible.	Pass

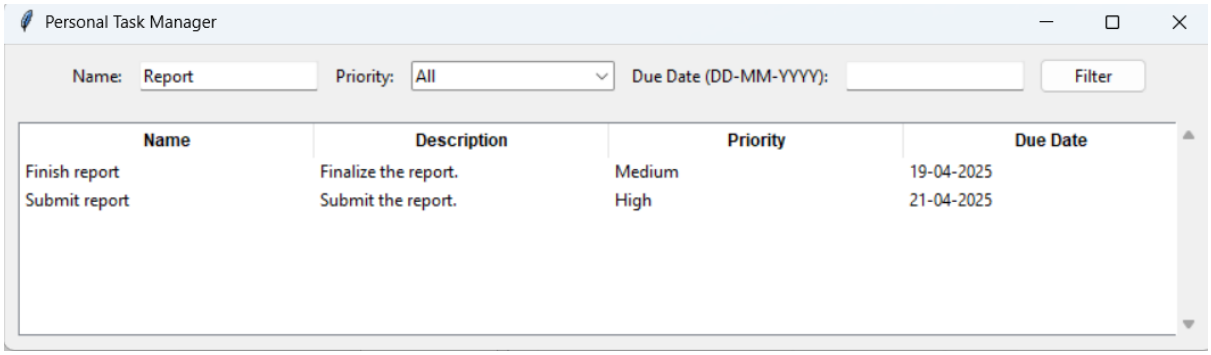


Figure 49: Test Case 30

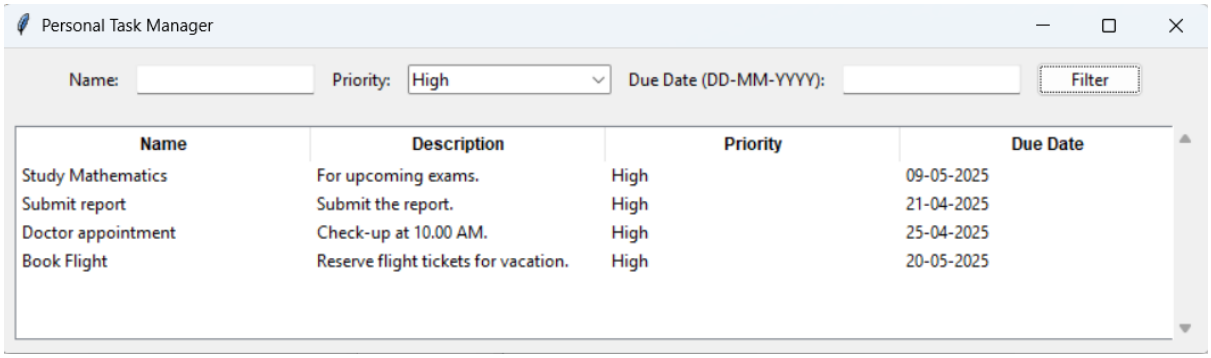


Figure 50: Test Case 31

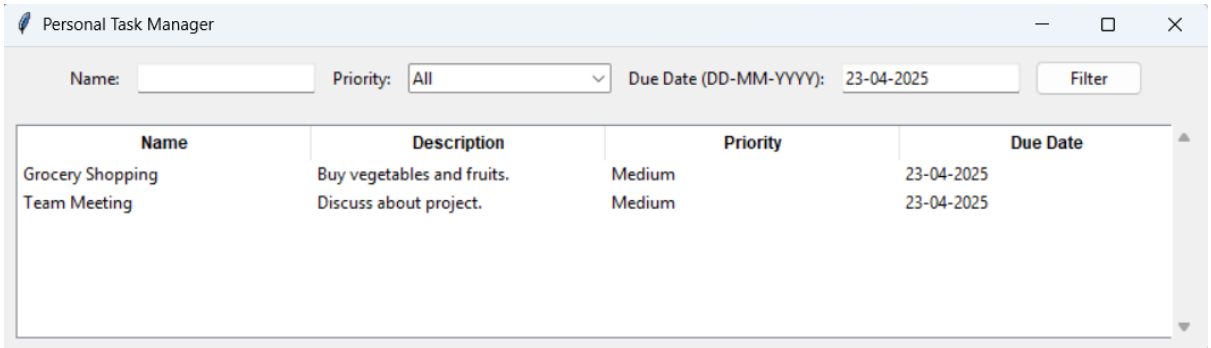


Figure 51: Test Case 32

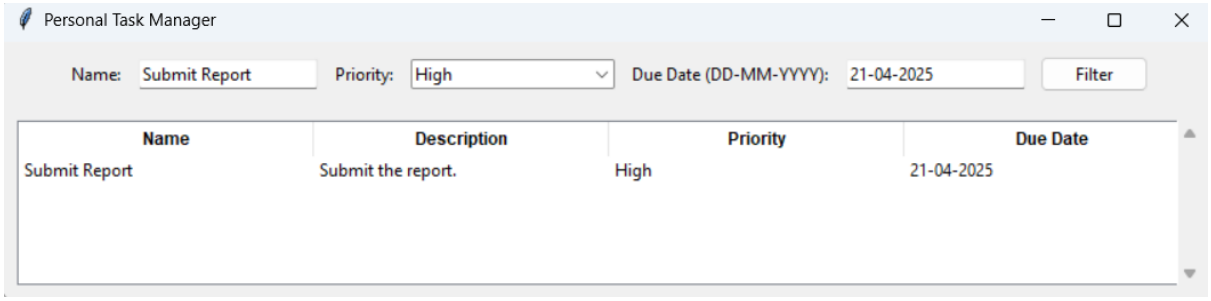


Figure 52: Test Case 33

Test Case: Sort Tasks

Task ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_34	Click the header "Name".	Sort data according to task name.	Sort data according to task name.	Pass
TC_35	Click the header "Description".	Sort data according to description.	Sort data according to description.	Pass
TC_36	Click the header "Priority".	Sort data according to priority (High, Medium, Low).	Sort data according to priority (High, Medium, Low).	Pass
TC_37	Click the header "Due Date".	Sort data according to date.	Sort data according to date.	Pass

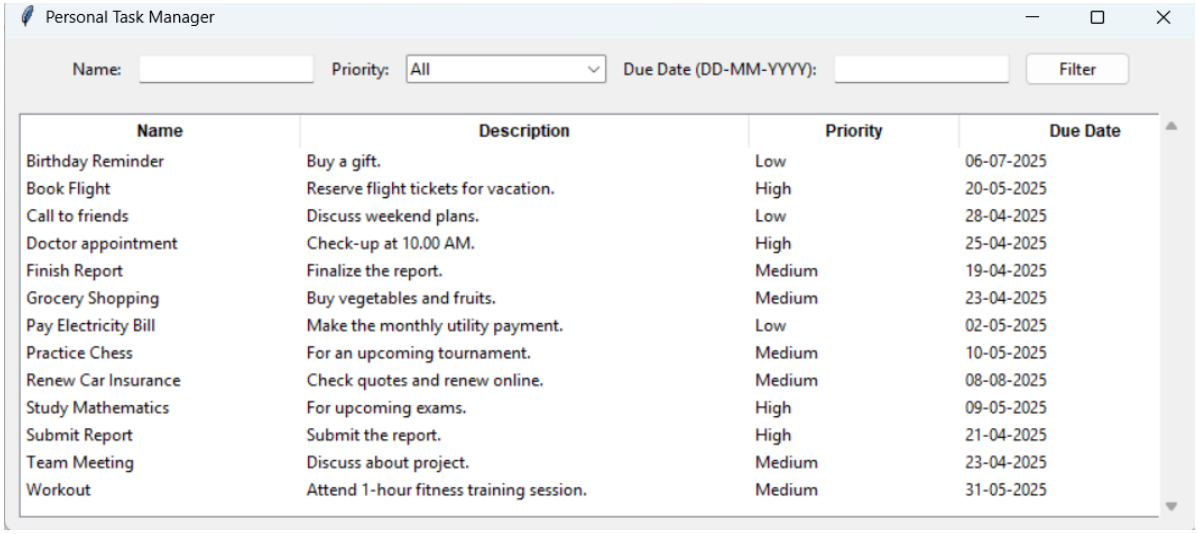


Figure 53: Test Case 34

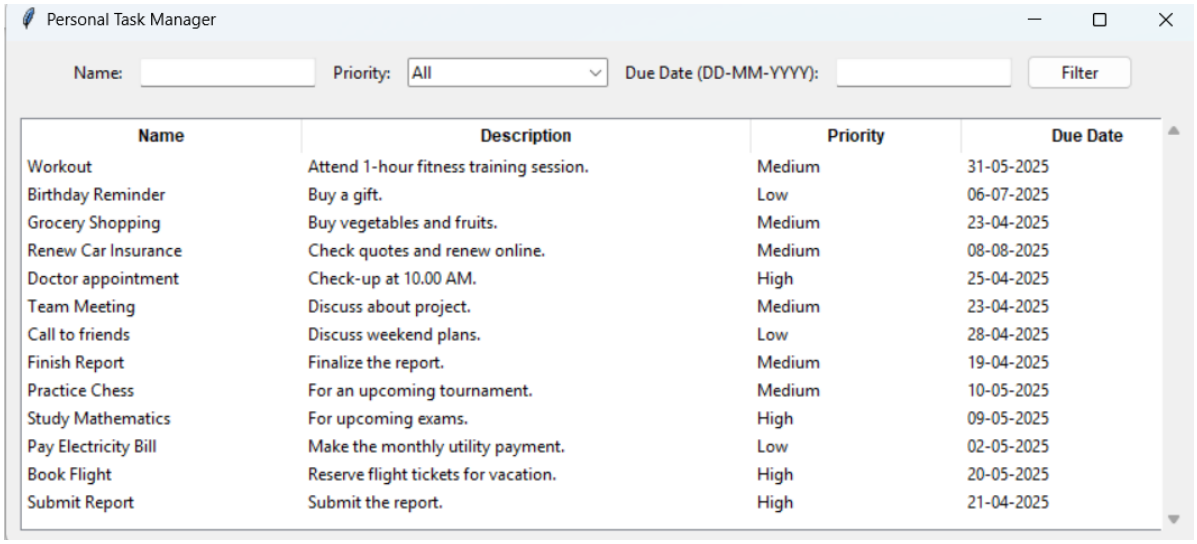
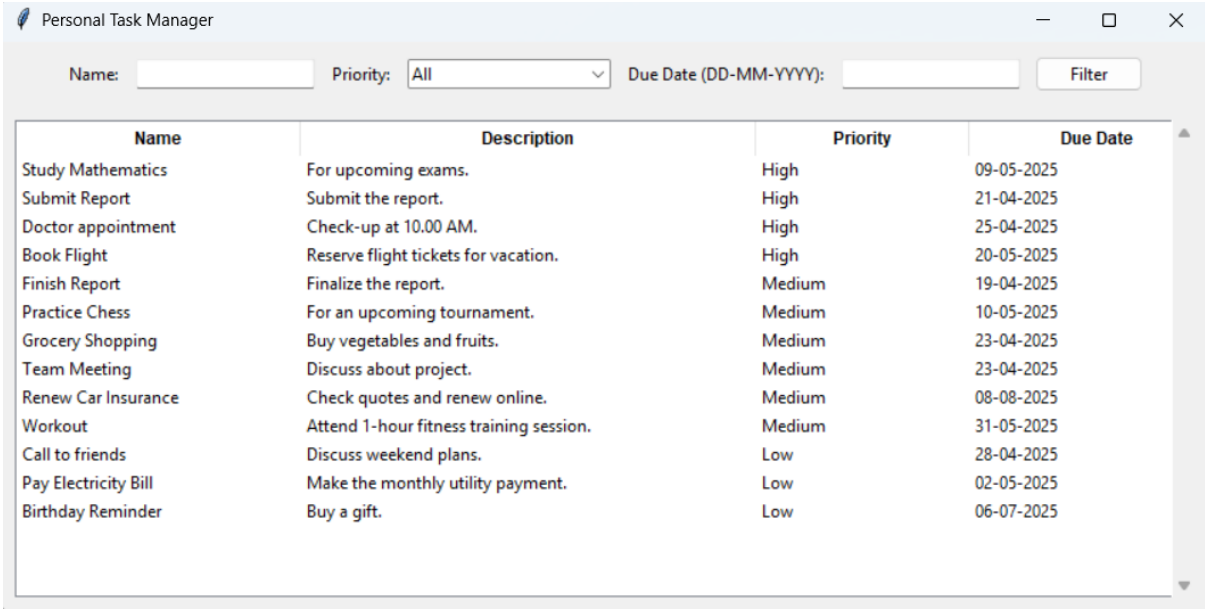
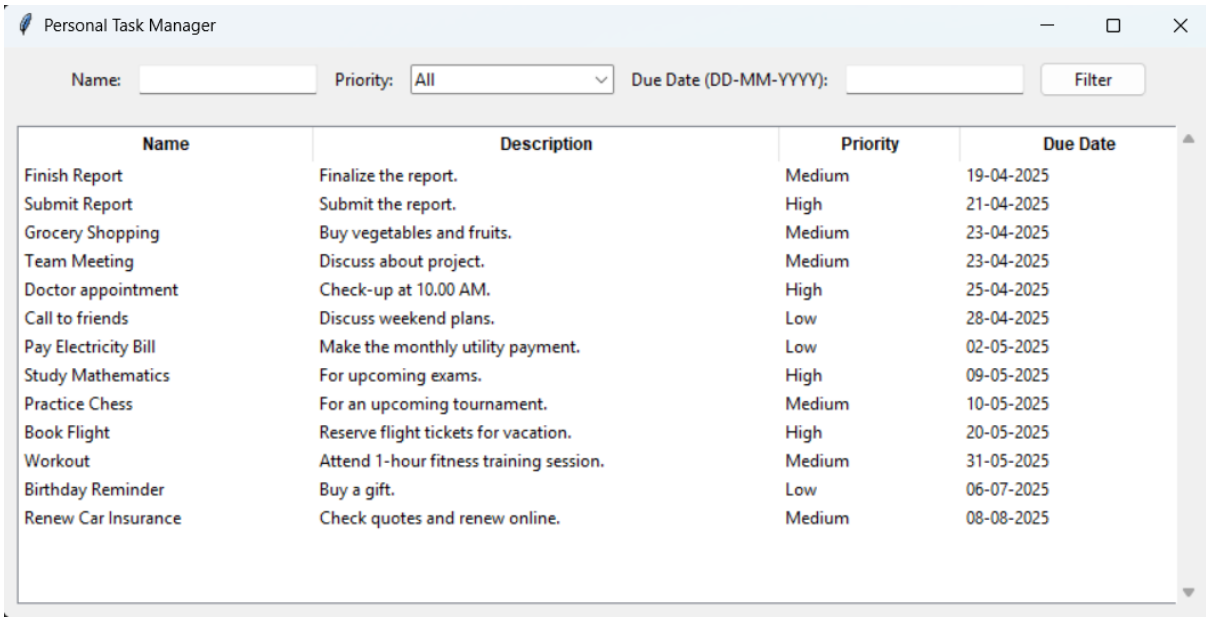


Figure 54: Test Case 35



Name	Description	Priority	Due Date
Study Mathematics	For upcoming exams.	High	09-05-2025
Submit Report	Submit the report.	High	21-04-2025
Doctor appointment	Check-up at 10.00 AM.	High	25-04-2025
Book Flight	Reserve flight tickets for vacation.	High	20-05-2025
Finish Report	Finalize the report.	Medium	19-04-2025
Practice Chess	For an upcoming tournament.	Medium	10-05-2025
Grocery Shopping	Buy vegetables and fruits.	Medium	23-04-2025
Team Meeting	Discuss about project.	Medium	23-04-2025
Renew Car Insurance	Check quotes and renew online.	Medium	08-08-2025
Workout	Attend 1-hour fitness training session.	Medium	31-05-2025
Call to friends	Discuss weekend plans.	Low	28-04-2025
Pay Electricity Bill	Make the monthly utility payment.	Low	02-05-2025
Birthday Reminder	Buy a gift.	Low	06-07-2025

Figure 55:Test Case 36



Name	Description	Priority	Due Date
Finish Report	Finalize the report.	Medium	19-04-2025
Submit Report	Submit the report.	High	21-04-2025
Grocery Shopping	Buy vegetables and fruits.	Medium	23-04-2025
Team Meeting	Discuss about project.	Medium	23-04-2025
Doctor appointment	Check-up at 10.00 AM.	High	25-04-2025
Call to friends	Discuss weekend plans.	Low	28-04-2025
Pay Electricity Bill	Make the monthly utility payment.	Low	02-05-2025
Study Mathematics	For upcoming exams.	High	09-05-2025
Practice Chess	For an upcoming tournament.	Medium	10-05-2025
Book Flight	Reserve flight tickets for vacation.	High	20-05-2025
Workout	Attend 1-hour fitness training session.	Medium	31-05-2025
Birthday Reminder	Buy a gift.	Low	06-07-2025
Renew Car Insurance	Check quotes and renew online.	Medium	08-08-2025

Figure 56:Test Case 37